

My Coach

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1. Introduction and Presentation of the Idea

1.1 Context and Objectives

In recent years, the demand for personalized fitness and nutrition solutions has surged, driven by a growing awareness of health and wellness. However, many individuals struggle to find a **single, accessible platform** that combines training programs, nutritional guidance, and community support. This project aims to address this gap by developing “**My Coach**”, a web application designed to **simplify fitness management** and provide users with a **comprehensive, interactive experience**.

The primary objective of “**My Coach**” is to offer a **user-friendly platform** where individuals can access personalized training programs, track their progress, and engage with a community of like-minded users. By integrating these features, the application seeks to **motivate users, enhance their fitness journey, and promote long-term health habits**.

1.2 The Original Idea

“**My Coach**” is a **web-based fitness platform** that provides users with tailored training programs, nutrition plans, and a supportive community. The application allows users to:

- Select and follow **personalized workout programs** based on their fitness goals and experience level.
- Access **customized nutrition plans** to complement their training.
- Engage with a **community** through challenges, leaderboards, and shared achievements.
- Purchase **fitness products and supplements** directly from the platform.

The platform is designed with a **clean, intuitive interface** to ensure ease of use, making it accessible to both beginners and experienced fitness enthusiasts.

1.3 Defense of the Idea

The development of “My Coach” is justified by several key factors:

- **Simplicity:** The platform consolidates multiple fitness tools into a **single, easy-to-use application**, eliminating the need for users to juggle multiple apps or services.
- **Personalization:** By offering tailored programs and nutrition plans, “My Coach” ensures that users receive **relevant, effective guidance** based on their individual needs.
- **Community Engagement:** The inclusion of a community feature fosters **motivation and accountability**, which are critical for long-term fitness success.
- **Educational Value:** The platform educates users about fitness and nutrition, empowering them to make **informed decisions** about their health.

In summary, “My Coach” is more than just a fitness app—it’s a **complete platform** that helps users reach their health and fitness goals.

2. Conception and Design

2.1 Introduction

At this point, we're reaching a milestone in the project elaboration process. In order to have a global and highlight the major outlines, we need to transform these requirements into diagrams using UML. These charts summarize the actions that our users can perform while using our application.

2.2 Main Use Case Diagram

The figure below shows the general use case diagram of our application. This illustration will help to have a global view on the board outlines of our system and the main actions are summarized into use cases. We find as agreed the main actors and their interactions with the service.

Table 1: Details of the main use case diagram

Title :	• Main Use Case Diagram
Actors :	• User
Precondition :	• Internet connection required
Description :	• The user can use the website to access various information regarding physical training • The system running in the background will import the data and trigger in specific cases.
Post conditions :	• Displayed interfaces

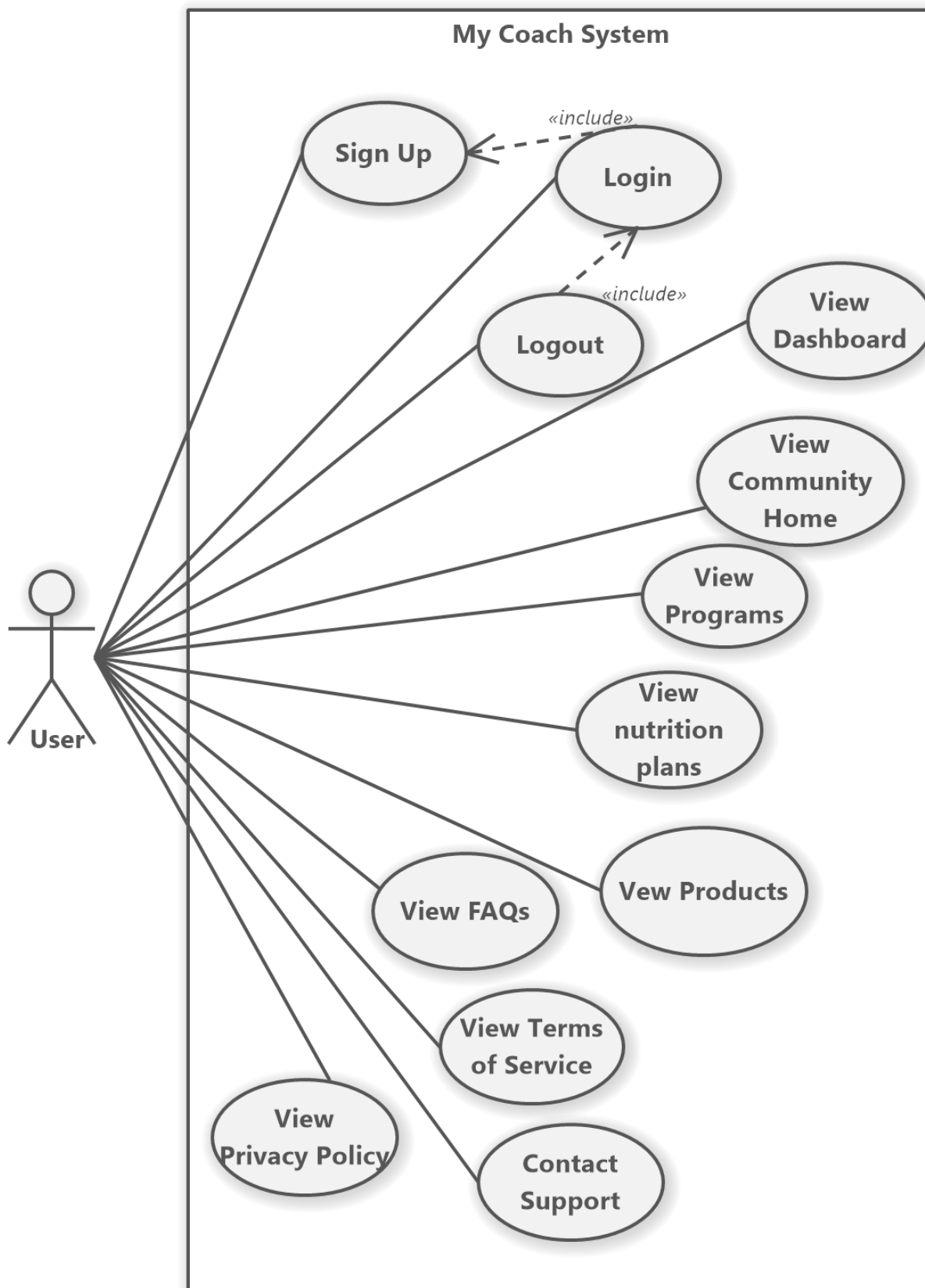


Figure 1: General Use Case

2.3 Detailed Use Cases

In this part of this chapter, we are going to give a closer look at the different use cases mentioned in the previous diagram. This will give us a more defined idea of the flow process for a given functionality that the user can execute.

2.3.1 Use case diagram for view programs

The tables below sum up the use case "View Programs" It describes the actor and the conditions of our process.

Table 2: Details of View Programs use case diagram

Title :	• View Programs use case diagram
Actors :	• User • The System
Precondition :	• Internet connection required • Authentication required
Description :	• The user will fill a form in which he will select his body biometrics. • Based on this information, the system will show recommended programs. • Among these programs, the user will select one of them.
Post conditions :	• Selected program

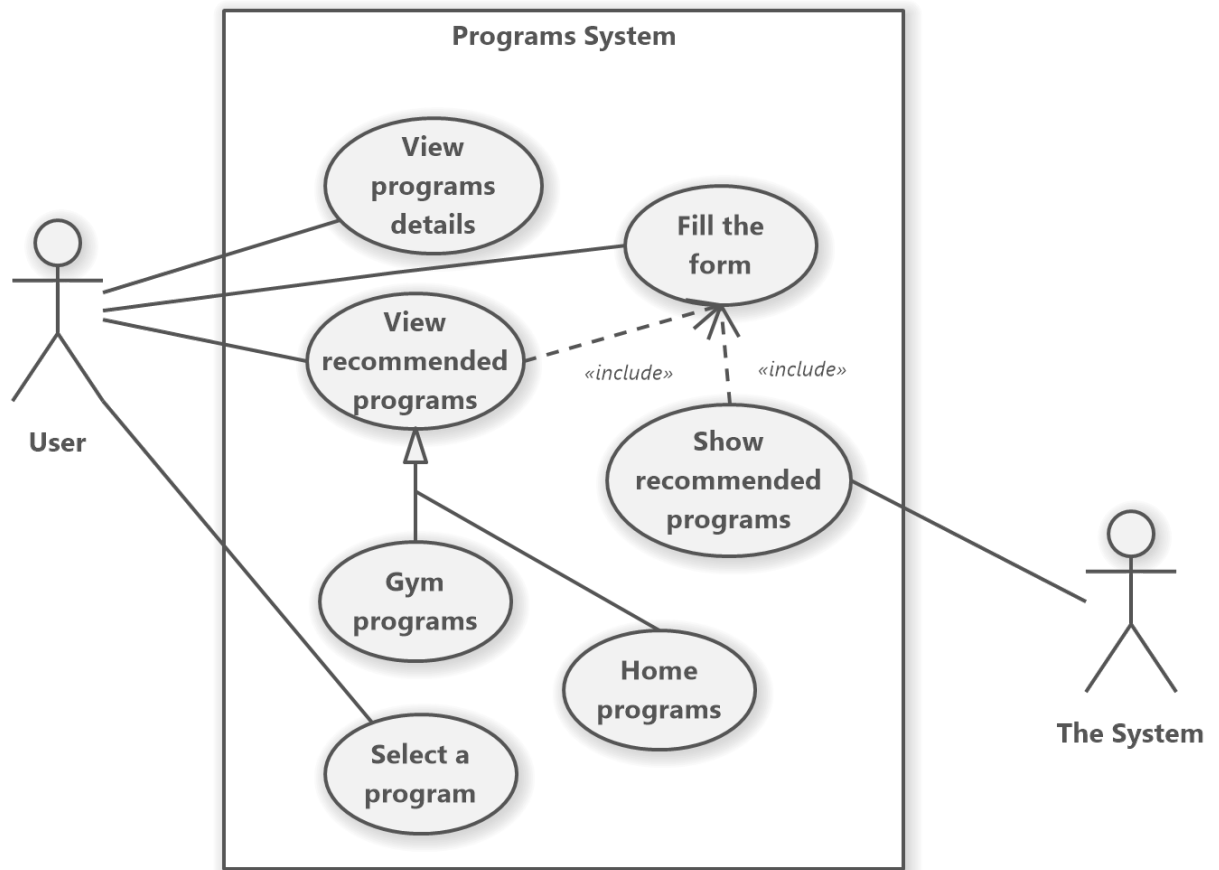


Figure 2: View Programs use case diagram

2.3.2 Use case diagram for View community home

The tables below sum up the use case "View Community home" It describes the actor and the conditions of our process.

Table 3: Details of View Community home use case diagram

Title :	• View Community Home use case diagram
Actors :	• User
Precondition :	• Internet connection required • Authentication required
Description :	• The user can browse the leaderboard, achievements, and stats • The user can join a challenge and earn points if it's completed.
Post conditions :	• Community leaderboard, recent achievements and stats interfaces displayed

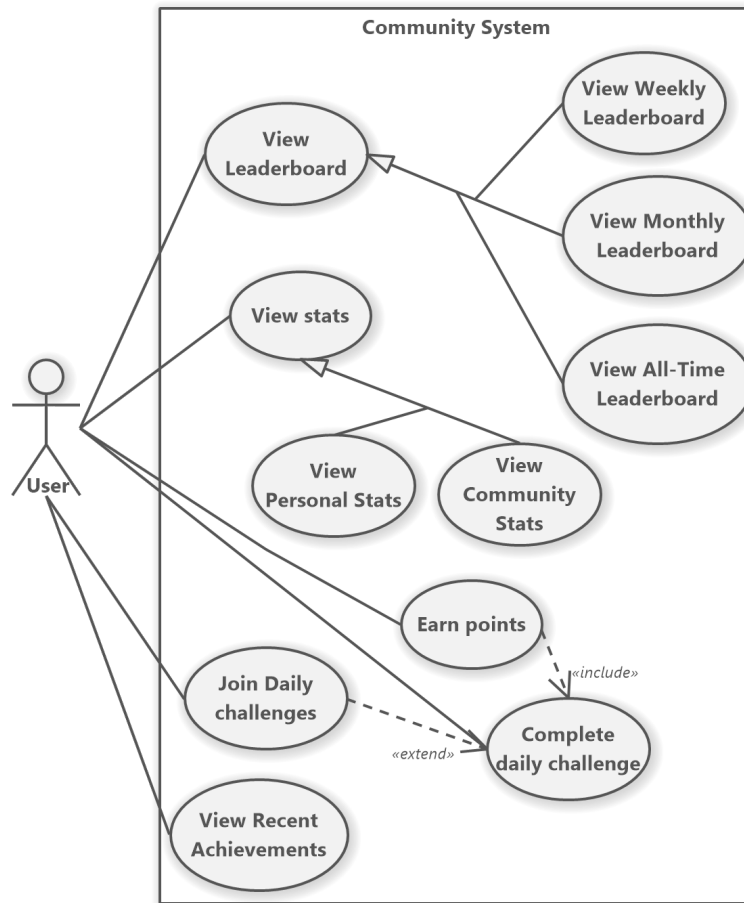


Figure 3: View Community use case diagram

2.3.3 Use case diagram for View Products

The tables below sum up the use case "View Products" It describes the actor and the conditions of our process.

Table 4: Details of View Products use case diagram

Title :	View Products use case diagram
Actors :	- User - Payment Gateway
Precondition :	- Internet connection required - Authentication required - Have a payment card
Description :	The user can buy sporting equipment and supplements.
Post conditions :	Product Purchased

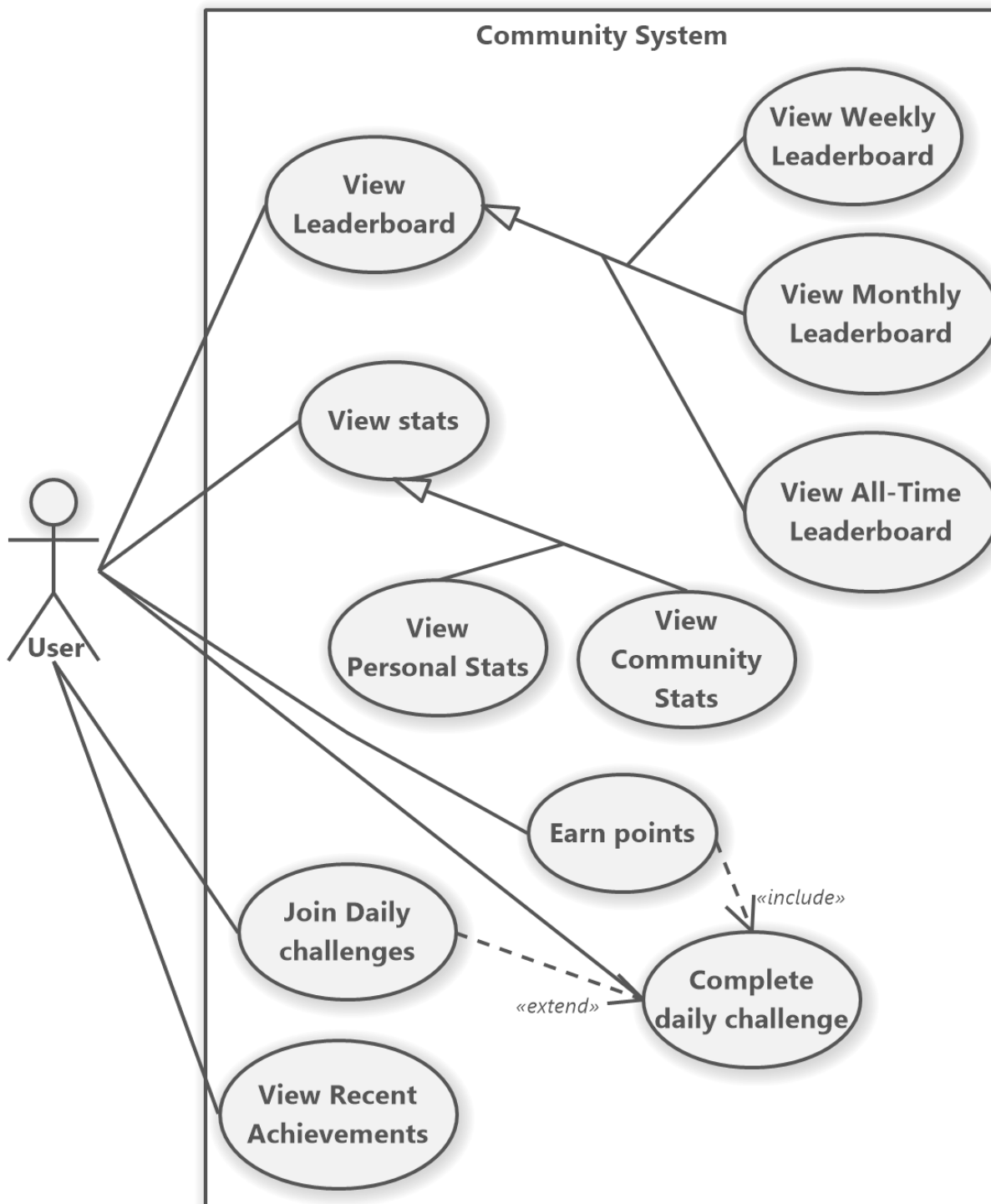


Figure 4: View Products use case diagram

2.4 Sequence Diagrams

2.4.1 Authentication

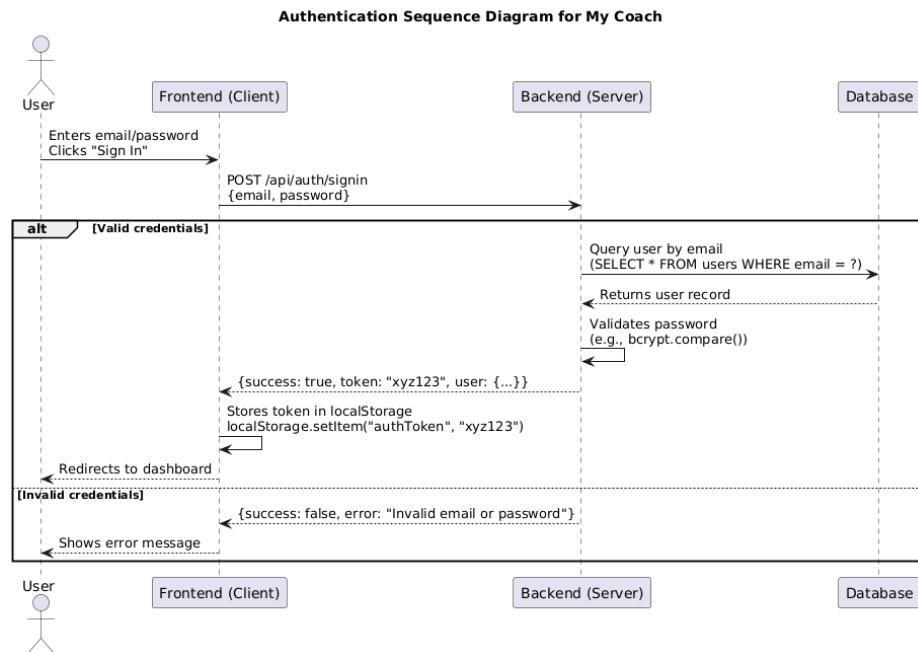


Figure 5: Sequence diagram for user authentication in "My Coach".

2.4.2 Choosing a Training Program

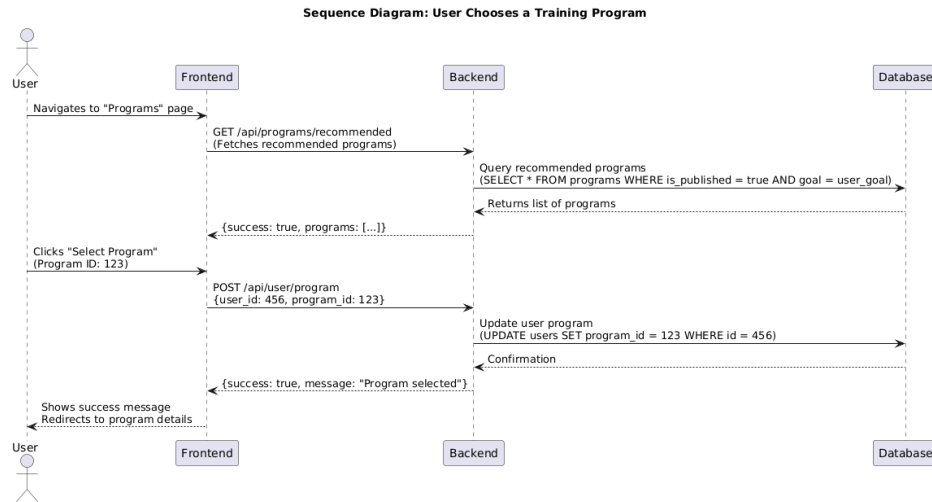


Figure 6: Sequence diagram for choosing a training program.

2.4.3 Choosing a Nutrition Plan

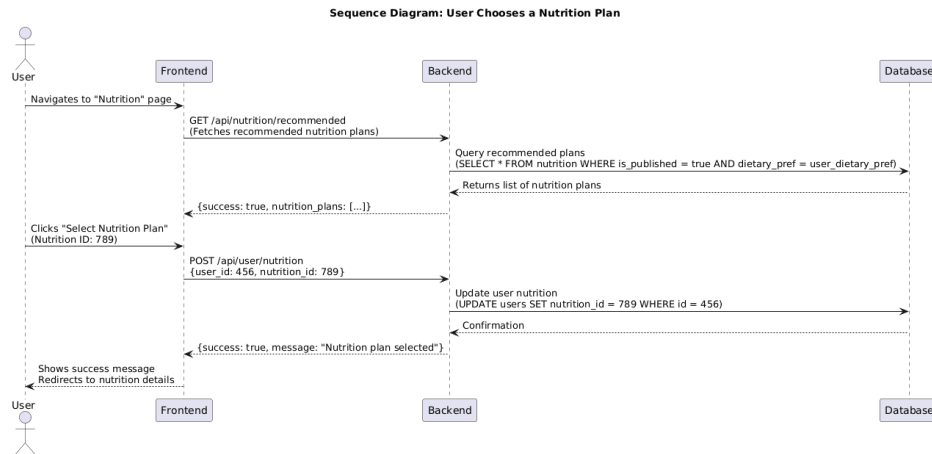


Figure 7: Sequence diagram for choosing a nutrition plan.

2.4.4 Filling the User Profile Form

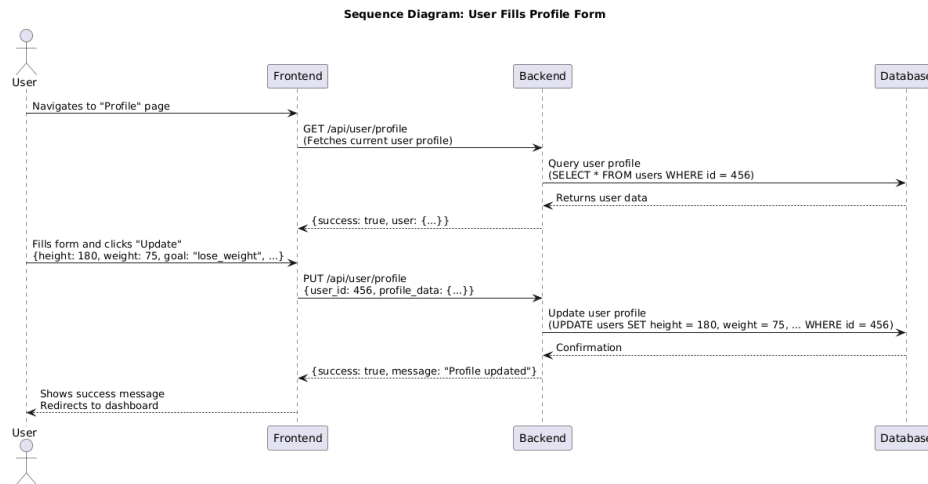


Figure 8: Sequence diagram for filling the user profile form.

2.5 Additional Diagrams

2.6 Technical Architecture (Mobile/Backend/Database)

The technical architecture is structured into four parts: the client, web server, application server, and database. On the client side we use HTML, CSS, and JavaScript, which run on the user's browser and are responsible for displaying the interface and sending HTTP/HTTPS requests according to user actions. The web server (e.g. Nginx) handles the requests and passes API calls to the backend. The application server is based on the Backend-as-a-Service Supabase, which implements the business logic executing methods and triggers written in PL/SQL in the SQL editor and exposes RESTful APIs automatically. The database layer, based on PostgreSQL, takes care of data storage, management, and integrity and, when needed, the execution of triggers and procedures. Data flows from the client to the web server, then to Supabase and the database, before returning the processed results back to the user.

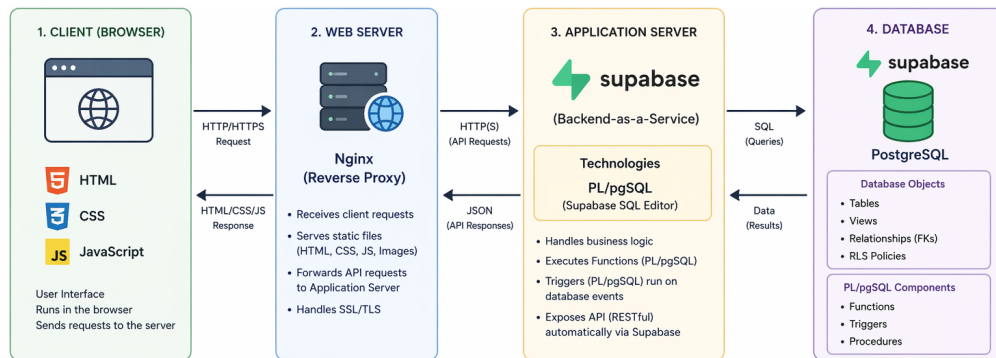


Figure 9: Architecture schema

2.7 Mockups

2.7.1 Main page

Landing Page It's the main entry point. There's a value proposition in the center and a simple "Hero" section. The ergonomic layout works well, creating a powerful focal point that immediately conveys the platform's objective. We simplified the navigation bar to only include necessary features to reduce "choice overload." We also positioned the profile symbol in the upper-right corner to match universal user mental models for easy account access..

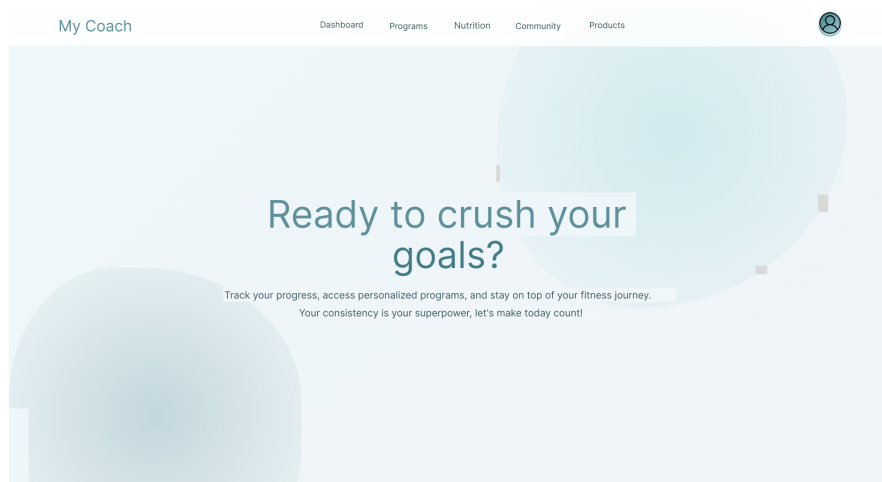


Figure 10: Main page Mockup

2.7.2 Features page

The “What We Offer” section showcases the main services provided by the platform in a card-based modular layout. The technique utilizes “Information Chunking”, which helps users quickly grasp a wide variety of products, including personalized programs and nutrition, without feeling cognitively exhausted. The great use of white space and consistent card padding, which makes the interface look professional and offers great readability, makes the interface feel open and friendly to novice users.

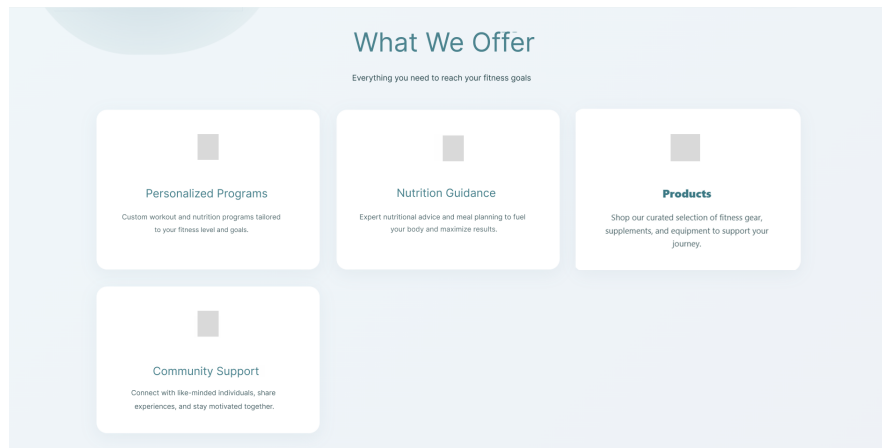


Figure 11: Features page Mockup

2.7.3 Sign Up page

The input fields on the Sign-up screen are in a logical order from basic identity (Username/Email) to security credentials. The linear flow reduces onboarding friction by aligning with user expectations. That same “Sign in” link at the bottom is also an important “navigation anchor” that makes it easy for visitors to change context if they already have an account. This improves the flow of the user journey.

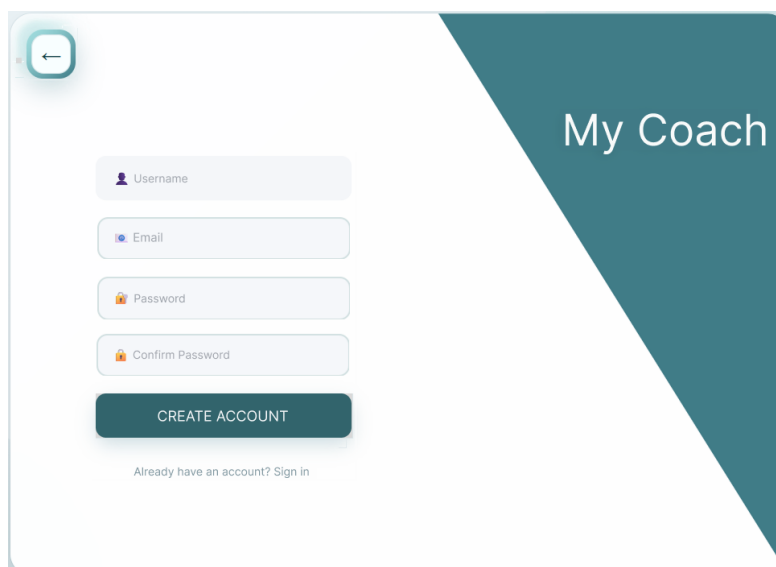


Figure 12: Sign Up page Mockup

3. Development

3.1 Chosen Technologies and Justifications

3.2 Code Organization

3.3 Implementation of Key Features

3.4 Data Management

4. Testing and Validation

4.1 Test Environment and Tools

4.2 Test Scenarios

5. Conclusion and Perspectives

5.1 Project Assessment

5.2 Global Difficulties and Learnings

5.3 Possible Evolutions

6. Appendices

6.1 Link to Git Repository

6.2 Supplementary Screenshots

6.3 Configuration Files